PROJECT DESIGN PHASE-II

Project Planning & Lifecycle Roadmap

Project: **Credit Card Fraud Detection System** • Developer: **Bhargavasai**

PRIMARY OBJECTIVE GOAL

Design, build, optimize, and deploy an end-to-end operational Machine Learning web application pipeline engineered specifically to classify real-time financial transaction streams into legitimate or fraudulent vectors.

01

Dataset Collection & Aggregation

Sourcing highly granular transactional records containing time matrices, numeric value properties, and anonymized credit dimensions.

02

Data Preprocessing & Data Cleaning

Executing full integrity passes including handling missing data items, removing duplicate instances, and scaling metrics uniformly via StandardScaler.

03

Exploratory Data Analysis (EDA)

Analyzing hidden feature structures and class imbalances through distribution histograms, scatter charts, and multi-variable correlation heatmaps.

04

Machine Learning Model Building

Training robust multi-algorithmic classification boundaries (Logistic Regression, Decision Trees, Random Forests, and XGBoost structures).

05

Flask Application Development

Constructing responsive HTML/CSS presentation wrappers and writing Flask API routing modules to load serialized pipelines via Joblib.

06

System Testing & Pipeline Evaluation

Auditing computational behaviors via classification reports, error logs, accuracy indicators, and confusion matrix testing arrays.

07

Production Cloud Deployment

Pushing version-controlled code frameworks onto GitHub repositories and establishing active live hosting URLs via the Render cloud environment.